Year 10 Higher

REVISION PACK

Algebra 2

Name: _____

Class: _____

Algebra 2 · Index laws with algebra (Lesson 7.1)

Worked example: simplifying with index laws

Simplify fully

(a) $(2x^3y)^5$ (b) $\dfrac{15a^6b^2}{5a^2b^4}$

(a)	**Step 1**	Number	$2^5 = 32$
	Step 2	Letters: multiply powers	$x^{15}, \quad y^5$
	Step 3	Answer	$32x^{15}y^5$
(b)	**Step 1**	Numbers	$15 \div 5 = 3$
	Step 2	Letters: subtract powers	$a^4, \quad b^{-2}$
	Step 3	Answer	$\frac{3a^4}{b^2}$

Practice: simplifying with index laws

1.1 Simplify fully

(a) $(2x^4y^2)^3$ (b) $\dfrac{18x^7y^3}{6x^2y^5}$

1.2 Simplify fully

(a) $(5a^2b^3)^2$ (b) $\dfrac{20a^4b}{4ab^3}$

1.3 Simplify fully

(a) $(3x^5y^2)^4$ (b) $\dfrac{24m^8n^3}{8m^3n^7}$

1.4 Simplify fully

(a) $(2a^3b^4)^3$ (b) $\dfrac{30a^5b^2}{6a^3b^6}$

1.5 Simplify fully

(a) $(4x^2y)^3$ (b) $\dfrac{21x^4y^2}{7xy^5}$

1.6 Simplify fully

(a) $(2m^5n^2)^4$ (b) $\dfrac{16p^6q}{4p^2q^4}$

Worked example: fractional powers with letters

Simplify fully $(8x^9y^3)^{\frac{2}{3}}$

Step 1	Number: root, then power	$(\sqrt[3]{8})^2 = 2^2 = 4$
Step 2	Letters: multiply the powers	$9 \times \frac{2}{3} = 6, \quad 3 \times \frac{2}{3} = 2$
Step 3	Answer	$4x^6y^2$

Practice: fractional powers with letters

1.7 Simplify fully $(125x^6y^9)^{\frac{2}{3}}$

1.8 Simplify fully $(64x^9y^3)^{\frac{2}{3}}$

1.9 Simplify fully $(25x^4y^8)^{\frac{3}{2}}$

1.10 Simplify fully $(8x^6y^{12})^{\frac{2}{3}}$

1.11 Simplify fully $(16x^8y^{12})^{\frac{3}{4}}$

1.12 Simplify fully $(1000x^3y^6)^{\frac{2}{3}}$

Worked example: simplifying algebraic fractions

Simplify fully $\dfrac{x^2+3x}{x^2-9}$

Step 1	Factorise the top	$x(x+3)$
Step 2	Factorise the bottom	$(x+3)(x-3)$
Step 3	Cancel $(x+3)$	$\frac{x}{x-3}$
Step 4	Answer	$\frac{x}{x-3}$

Practice: simplifying algebraic fractions

2.1 Simplify fully $\dfrac{x^2-5x}{x^2-25}$

2.2 Simplify fully $\dfrac{x^2+2x}{x^2-4}$

2.3 Simplify fully $\dfrac{x^2-36}{3x^2+18x}$

2.4 Simplify fully $\dfrac{x^2-7x}{x^2-49}$

2.5 Simplify fully $\dfrac{x^2+6x}{x^2-36}$

2.6 Simplify fully $\dfrac{x^2-1}{2x^2+2x}$

Worked example: fractions with a quadratic

Simplify fully $\dfrac{x^2 + x - 12}{x^2 - 9}$

Step 1	Factorise the top	$(x + 4)(x - 3)$
Step 2	Factorise the bottom	$(x + 3)(x - 3)$
Step 3	Cancel $(x - 3)$	$\dfrac{x + 4}{x + 3}$
Step 4	Answer	$\dfrac{x + 4}{x + 3}$

Practice: fractions with a quadratic

2.7 Simplify fully $\dfrac{x^2 + 3x - 10}{x^2 - 4}$

2.8 Simplify fully $\dfrac{x^2 - x - 12}{x^2 - 16}$

2.9 Simplify fully $\dfrac{x^2 + 5x + 6}{x^2 - 9}$

2.10 Simplify fully $\dfrac{x^2 - 25}{x^2 - 3x - 10}$

2.11 Simplify fully $\dfrac{x^2 + 4x - 5}{x^2 - 1}$

2.12 Simplify fully $\dfrac{x^2 - 2x - 8}{x^2 - 4x}$

Worked example: dividing algebraic fractions

Simplify fully $\dfrac{x+2}{4} \div \dfrac{x^2-4}{8}$

Step 1 Flip and multiply	$\dfrac{x+2}{4} \times \dfrac{8}{x^2-4}$
Step 2 Factorise	$\dfrac{x+2}{4} \times \dfrac{8}{(x+2)(x-2)}$
Step 3 Cancel	$(x+2), \quad 8 \div 4 = 2$
Step 4 Answer	$\dfrac{2}{x-2}$

Practice: dividing algebraic fractions

2.13 Simplify fully $\dfrac{x+3}{5} \div \dfrac{x^2-9}{10}$

2.14 Simplify fully $\dfrac{x-4}{2} \div \dfrac{x^2-16}{6}$

2.15 Simplify fully $\dfrac{x+1}{6} \div \dfrac{x^2-1}{3}$

2.16 Simplify fully $\dfrac{x+6}{4} \div \dfrac{x^2+6x}{12}$

2.17 Simplify fully $\dfrac{x-7}{3} \div \dfrac{x^2-49}{9}$

2.18 Simplify fully $\dfrac{x^2-4}{5} \div \dfrac{x+2}{10}$

Algebra 2 · Equations with powers and roots (Lesson 7.3)

Worked example: solving equations with roots and squares

Solve

(a) $\sqrt{3x+1}=4$ (b) $2x^2+3=35$

(a)	**Step 1**	Square both sides	$3x+1=16$
	Step 2	Subtract 1	$3x=15$
	Step 3	Divide by 3	$x=5$
	Step 4	Answer	$x=5$
(b)	**Step 1**	Subtract 3	$2x^2=32$
	Step 2	Divide by 2	$x^2=16$
	Step 3	Square root	$x=4$ or $x=-4$
	Step 4	Answer	$x=\pm 4$

Practice: solving equations with roots and squares

3.1 Solve

(a) $\sqrt{2x+5}=3$ (b) $2x^2-5=45$

3.2 Solve

(a) $\sqrt{4x+1}=5$ (b) $4x^2+1=37$

3.3 Solve

(a) $\sqrt{5x-4}=6$ (b) $3x^2-12=15$

3.4 Solve

(a) $\sqrt{2x+1}=5$ (b) $5x^2-3=77$

3.5 Solve

(a) $\sqrt{6x-2}=4$ (b) $2x^2+7=57$

3.6 Solve

(a) $\sqrt{3x-5}=2$ (b) $4x^2-9=27$

Algebra 2 · Equations with powers and roots (Lesson 7.3)

Worked example: solving with a bracket squared

Solve $(x-3)^2 = 25$

Step 1	Square root: + and −	$x - 3 = \pm 5$
Step 2	Add 3	$x = 3 + 5$ or $x = 3 - 5$
Step 3	Answer	$x = 8$ or $x = -2$

Practice: solving with a bracket squared

3.7 Solve $(x+2)^2 = 49$

3.8 Solve $(x-5)^2 = 36$

3.9 Solve $(x+6)^2 = 16$

3.10 Solve $(x-1)^2 = 64$

3.11 Solve $(x+3)^2 = 100$

3.12 Solve $(x-7)^2 = 9$

Algebra 2 · Equations with powers and roots (Lesson 7.3)

Worked example: cube roots and square roots

A cube has a volume of 400 cm³. Work out the length of one edge. Give your answer correct to 3 significant figures.

Step 1	Volume = edge³	$e^3 = 400$
Step 2	Cube root	$e = \sqrt[3]{400} = 7.368\ldots$
Step 3	Round to 3 s.f.	7.37
Step 4	Answer	$7.37\ \text{cm}$

Practice: cube roots and square roots

3.13 A cube has a volume of 150 cm³. Work out the length of one edge of the cube. Give your answer correct to 3 significant figures.

3.14 A square has an area of 90 cm². Work out the length of one side of the square. Give your answer correct to 3 significant figures.

3.15 A cube has a volume of 500 cm³. Work out the length of one edge of the cube. Give your answer correct to 3 significant figures.

3.16 A square has an area of 45 cm². Work out the length of one side of the square. Give your answer correct to 3 significant figures.

3.17 A cube has a volume of 80 cm³. Work out the length of one edge of the cube. Give your answer correct to 3 significant figures.

3.18 A cube has a volume of 1200 cm³. Work out the length of one edge of the cube. Give your answer correct to 3 significant figures.